

Ravenwood Veterinary Clinic

2015 Kennel Management

I. Record Keeping

1. Medical
 - a. Vaccination chart
 - b. Deworming chart
 - c. Female heat cycle chart
 - d. Individual - lameness, major surgery, etc.
2. Pedigrees on each animal
3. Mileage log in training
4. "Weekly Minder" - to keep track of breeding, weaning, when dogs are sick, etc. The above is a list we organized from experience. This insures you have the information, in writing, that your veterinarian needs to help you trouble shoot specific problems.

II. Vaccinations

1. Miscellaneous notes on vaccinations:
 - a. Dogs can be vaccinated after training for the day, but not immediately before running. Body temperatures over 104° F can inactivate the vaccines before they are absorbed.
 - b. Do not vaccinate pregnant females. Little information is available on the affects of modified live vaccines on fetal development.
 - c. Booster vaccines should not be given any closer than 10 days apart. The body's normal immune response can inactivate the second vaccine during this period.
 - d. I prefer you vaccinate in the fall since the dogs have been resting and their immune systems are at their peak. This will ensure highest vaccine titers during the winter months.
 - e. There are a variety of thoughts on the vaccine frontier and each year we receive more information. Most of the upheaval in vaccine considerations has come from speculation that cats who are genetically predisposed to a type of subcutaneous tumor may be stimulated to grow these tumors in chronic vaccine reaction sites. There has not been any evidence of this in dogs; however, this has prompted the universities to review the vaccine protocols. The heart of the confusion is vaccine companies are only required to show that their vaccines will protect a majority of vaccinated dogs for the time interval they state on their vaccines. This means no one really knows how long the vaccines will protect the individual animal and most company's only license for 12 months. Here at RVC we follow the guidelines laid out by the AAHA (American Animal Hospital Association) vaccine committee in program for core vaccine recommendations. These are reviewed by a panel of specialist every other year.
2. Competitive dogs, like competitive human runners, are training to a level that creates some immunosuppression. This means that the dogs are more susceptible to "bugs" and infections. It also means that an individual dog may not have as high a response to the vaccine as expected. (Thus, the reason for vaccinating in the fall before heavy training starts!). Because our sled dogs are exposed to a lot of different animals from all over the country the risk of exposure to these viruses is high. At this point in time the American Animal Hospital Association core vaccine guidelines recommend three year programs for dogs that have had a

solid initial series of vaccines. One exception would be breeding females. Since you want high antibody titers in the milk from the female, they should stay on annual combinations. Older sled dogs that are not traveling as much are good three year candidates and many events and kennels are moving to the three year recommendation for racing dogs as well. You will want to make sure you have maximized your dogs' health in other areas if you are going to the three year program.

3. On a similar line of thinking, we want to be sure our puppies are protected, but we don't want to add stress by over-vaccinating (i.e. you can't overcome parasite, poor feeding, or management problems by giving more vaccines). In light of the newer, low passage-high titer Parvo vaccines we are making the following recommendations:

Puppies (assuming dam is current on her vaccines at the time of breeding)

6 weeks - DA₂PV-CPV (Combo distemper/parvo/adenovirus)

10 weeks - DA₂PV-CPV

14 weeks - DA₂PV-CPV

Have a veterinarian vaccinate against rabies anytime after 12 weeks, as required by state and local laws.

4. Adults

- a. DA₂PV-CPV - annually, preferably in August or September, unless you want to go to the three year program after 2 years of age (puppy series and one annual).
- b. Rabies (administered by a veterinarian) - initially after 12 weeks of age, and repeated one year later. Subsequent rabies vaccinations are good for three years in the State of Alaska.
- c. Bordetella (Kennel Cough) - I now recommend the oral vaccine. To maximize protection, give in November and again two weeks before your major race. Do not run the dog for 24 hours after the oral vaccine. Some people have given the injectable vaccine two weeks after the intranasal for added protection. The injectable vaccine needs to be boosted two weeks after the first injection. If an outbreak of any respiratory infection occurs in your area, it helps to booster your non-symptomatic dogs immediately with the oral. At best, this respiratory vaccine is about 20% effective.
- d. Influenza vaccine – as of 6/2015 there has not been a documented case of canine influenza in the State of Alaska. Considering our high rate of traveling dogs, it is only a matter of time. This vaccine needs a primary and then 2 to 6 weeks later a booster to be protective. At this time we are only giving these vaccines on special request.

III. Worm Control

1. Types of worms:
 - a. "Rounds" (toxocara species, hooks, whips) – fecal-oral transmission, or nursing on infected mothers.
 - b. Tapeworms (Taenia, Echinococcus)- from fleas, raw meat, voles, and fish
2. Damage caused by:
 - a. Migration of larva through the liver and lungs causing scarring
 - b. Competition for calories in small intestines (creates a thin, unkempt appearance)
 - c. Indigestion from presence in small intestines creates stress diarrhea
3. Open sores in intestine allows other viral, bacterial, and parasitic agents

into the body.

4. Humans can be infected with "round" and tapeworm parasites. Small children are especially at risk from diminished hygiene awareness. There have been cases of migrating roundworm larva causing blindness in humans. Transmission is through ingesting eggs (fecal/oral) or picking up larva while walking barefoot. Both can be prevented with good hygiene.

5. Prevention:

- a. Deworm puppies with a thorough schedule.
- b. Deworm entire kennel regularly.
- c. Have stool samples checked at least once a year. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- d. CLEAN YARD AT LEAST ONCE A DAY to prevent larvae and eggs from becoming infective.
- e. Don't feed raw game meat.
- f. Move where there aren't any fleas.
- g. Get ready to deal with ticks. Ticks have been documented over wintering in both the Kenai and Fairbanks areas.

6. Types of dewormers:

- a. Fenbendazole (Panacur) - kills some migrating larvae as well as adult worms, effective against rounds, whips, hooks, and certain tapes (not the flea tapes), **MUST** be given daily for three days in a row. While not FDA approved for giardia, it does appear to show promise in treating giardia infections if given for 5 or more consecutive days.
- b. Piperazine (numerous brand names) - only kills roundworms, one dose, repeat in three weeks.
- c. Pyrantel Pamoate (Nemex, Strongid T) - kills only rounds and hookworms, one dose, repeat in three weeks.
- d. Praziquantel (Droncit) or Epsiprantel (Cestex) - kills only tapeworms. This drug is more toxic than previously mentioned dewormers. Research has not established use in pregnant or breeding animals.
- e. Praziquantel/Febantel/Pyrantel (Drontal-Plus)- a broad-spectrum product that eliminates roundworm species and both species of tapeworms in a single dose tablet.
- f. You should check with your veterinarian on ages of animals and doses of all dewormers since some can be toxic.
- g. Many of the spot-on treatments for fleas/ticks/lice/heartworm will treat some internal parasites as well.

7. Puppies:

Panacur to dam (for best elimination the specialists recommended every day from day 45 to day 63. There is evidence that treating day 45, 46, 47 will help decrease the overall amount of transplacental spread of the feti).

- a. Panacur to dam at day 45, 46, 47 of pregnancy to whelping.
- b. Panacur at two weeks of age (deworm dam each time as long as she is with pups)
- c. Pyrantel P. at five weeks of age

- d. Pyrantel P. at seven weeks of age
- e. Pyrantel P. at nine weeks of age
- f. Pyrantel P. at 11 weeks of age
- g. Pyrantel P. at 13 weeks of age
- h. Pyrantel P. at 15 weeks of age
- i. Panacur three consecutive days when put out of puppy pens
- j. Fecal tests should be run at three, six, and 21 weeks of age to evaluate your deworming success and check for other parasites such as coccidia and giardia.

7. Adults

- a. *Small yards* - one to four dogs with no litters on the ground. Treat once a year with 3 days of Panacur or one dose of Pyrantel P followed by a second dose three weeks later. Run fecal tests every 6 months. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- b. *Medium and Large Yards* - five or more dogs with no more than one litter per year, or any kennel that is racing at the local tracks. Treat all dogs three times a year (September, January, and March). Panacur should be one given at one of those times and Pyrantel P. the other two. Run fecal tests every 6 months. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- c. *Other situations* - five or more dogs with multiple litters each year, and any kennel that is racing at both local and state wide tracks. Treat five times a year (April, August, October, December and February). Again, at least one treatment with Panacur and the others with Pyrantel P. Run fecal tests every 6 months. We recommend samples representing 20% of your kennel at a time i.e., 0 to 10 dogs - 2 samples, 10 to 20 - 4 samples, etc.
- d. *Mega yards* - lots of dogs and lots of puppies. Given the high, essentially uncontrollable incidence of infection in the soil, it is best to deworm all dogs once a month. Deworm at least twice with Panacur and the rest with Pyrantel P.
- e. Consider Drontal-Plus deworming once a year in any kennel feeding a meat diet. Though higher in price per dog, this product effectively kills all species of tapeworms in one dose, including Echinococcus. Iditarod requires all competitors to deworm with Drontal-Plus due to the potential for human infection from this parasite. Kennels north of Talkeetna and anyone feeding raw game meat should consider annual tape deworming for Enchinococcus.

IV. Other Parasites to Watch for:

Coccidia, giardia, lice, mites, ticks, fleas, heartworm.

V. Nutrition

Rules of thumb

- a. Use a proven diet (one that someone else has been using for a while with success) or go straight commercial to prevent nutritional deficiencies or toxicities.
- b. If formulating your own diet, work with a veterinarian or a nutritionist.
- c. Make any changes in diet gradually, change only one ingredient at a time, and give 4 to 6 weeks to see any difference (skin and coat can take up to eight weeks to show the effects of diet changes).
- d. If feeding more than 25% meat (this is straight muscle meat, not necessarily commercial meat mixtures) in a diet, be sure to calculate the calcium: phosphorus

ratio (preferably 2 calcium to 1 phosphorus). Ensure that the levels are within a safe range for the dogs' age by consulting with the veterinary clinic or finding an AAFCO (American Animal Feed Control Official) table for the daily requirements.

- e. Puppies (dogs ten months and younger) should be on straight commercial puppy food. We have found a number of race diets are too "hot" and the calcium: phosphorus ratio is not balanced for proper puppy bone growth.
- f. You will get into trouble by heavily supplementing with any one vitamin or mineral. If you want to supplement, get hold of a current NRC table and calculate how much is in the total diet.
- g. Look for articles written by Dr. Arleigh Reynolds on sled dog nutrition.
- h. Because vitamins/supplements are not regulated by FDA you have to be cautious of your source. One recent study found out of 10 samples of glucosamine tested only 3 actually had what and the amount stated on the packaging. The following is an independent laboratory that you can join and have access to products that have been tested and proven to have what they label. ConsumerLab.com

VI. Miscellaneous

1. If you do have a fatality with a puppy, I highly recommend you aggressively pursue an accurate answer. The cost of a necropsy with viral, parasitic, bacterial, and histological exams may appear expensive at the time; however, it won't be nearly as expensive as losing all of your pups. Owners are often surprised by how correctable the problem can be.
2. Nails - trim nails before the start of cart training to prevent breaks. Leave them sufficiently long to account for hard surface wear. Once the snow falls, check nails the first of each month you are running. Do not trim nails the day before a major race! If your race surface includes snow on pavement, do not leave the nails too long but do not cut them any closer than two weeks to the event.
3. Safety - always carry a knife, extra double snaps, and common sense when running dogs. Be sure you can safely carry a dog in your sled. Remember, you put the dog in the situation and it is up to you to safely get him out.
4. Ask a lot of questions all the time!
5. Don't cuss out your veterinarian if you don't follow their instructions.
6. "The one important thing I have learned over the years is the difference between taking one's work seriously and taking one's self seriously. The first is imperative and the second is disastrous." – Margot Fonteyn (1919-1991).

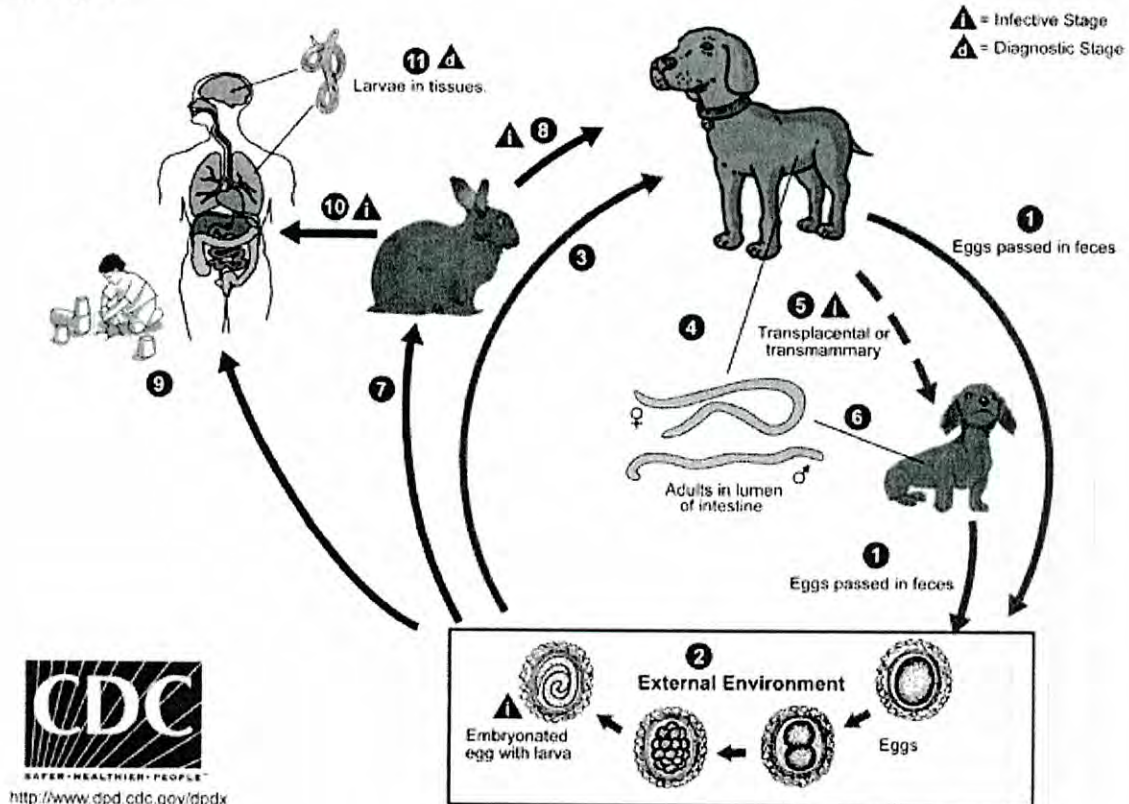
Toxocara canis – Life Cycle and how it transmits in the environment (and to people).

Biology

Causal Agents:

Toxocariasis is caused by larvae of *Toxocara canis* (dog roundworm) and less frequently of *T. cati* (cat roundworm), two nematode parasites of animals.

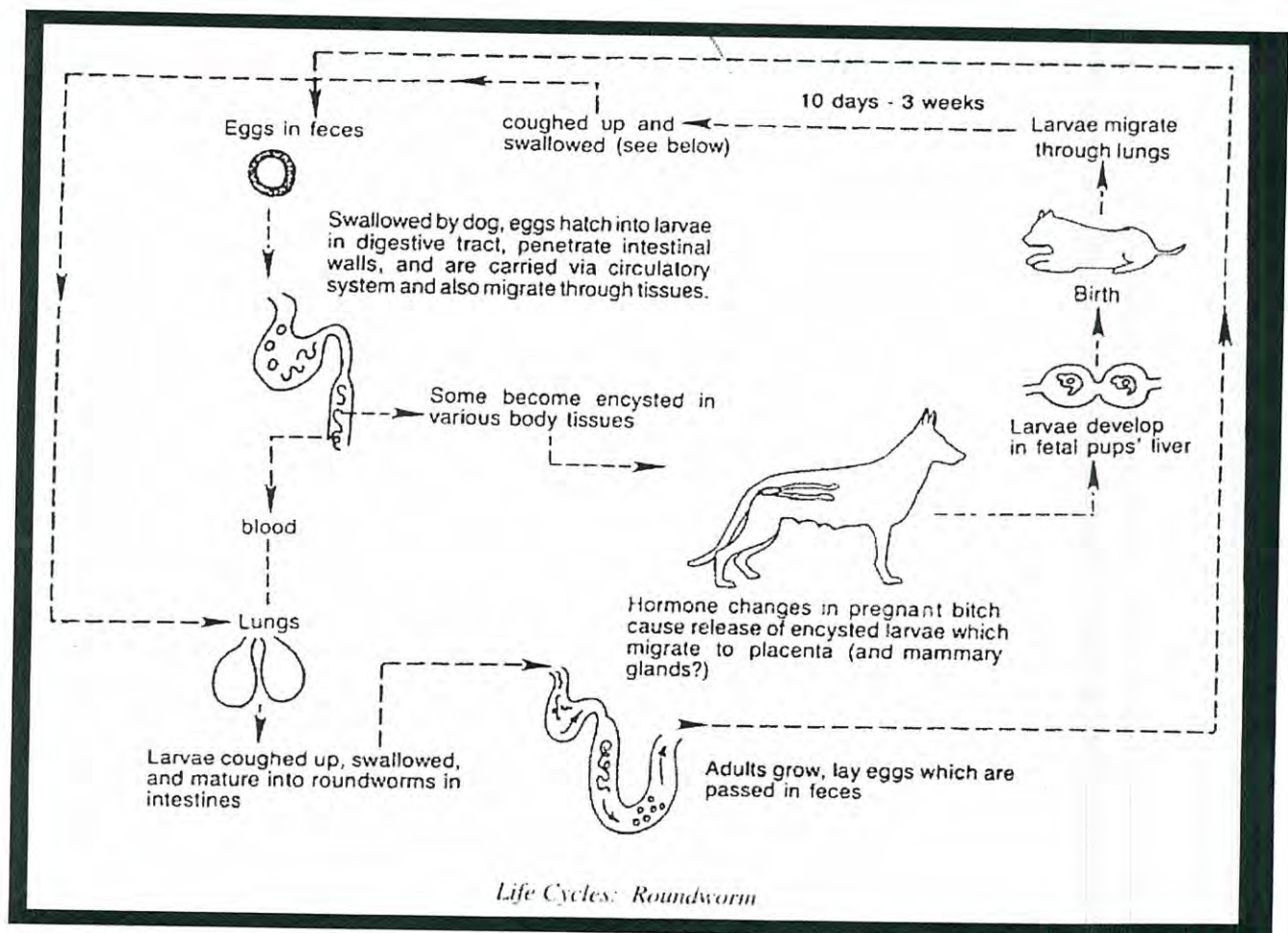
Life Cycle:



<http://www.dpd.cdc.gov/dpdx>

Toxacara canis life cycle in the body of the dog:

With a heavy roundworm parasite load it is possible for a three month old puppy to be suffering from liver, lung, and intestinal damage. Not to mention all the encysted parasites that could end up in the muscle tissue, which is suppose to be moving the dog!



Dog Mushers' Emergency Kit Continued

Use	Administration	Instructions
Eye injuries	For eyes only	A bland trithalamic Eye Ointment - Apply to injured eye every 4 hours until examined by a Veterinarian.
Ear infections	For ears only	Choose a mild ear or otic cleanser. Epi-Otic advanced is the only one approved in ears with ruptured ear drums.
Mild stress diarrhea, or after eating sharp objects or garbage	Oral	Metamucil (plain) - 1 tsp. powder or 1 wafer per 30 lbs of dog. (Or pumpkin or beet pulp.) Mix powder with moist food or cottage cheese. Give 4 to 6 hours before a run to help with chronic stress diarrhea.

- ◆ Minimize all licking of ointments and creams by the dog. Many foot ointments contain zinc oxide, which if ingested, in volume and/or over a period can cause serious medical problems.

Bandage Material and Other Useful Items to Have on Hand

- ◆ Telfa pads - apply over open or draining wounds before bandaging
- ◆ Gauze brown cling - use to hold telfa pad in place
- ◆ Vet wrap - to support or protect affected areas
- ◆ Neoprene wraps with "mineral ice"
- ◆ Dental acrylic with gel caps or "Soft Paws" nail caps
(Sterilize nail tip before applying and always remove immediately after race)
- ◆ Bandage scissors
- ◆ Nail trimmers
- ◆ At least one good leash
- ◆ Muzzle
- ◆ Extra booties

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2015
Controlling Stress Diarrhea in Working Dogs

1. A complete and thorough parasite control program should be in place to help rule out parasitism. Even so, multiple fecal samples may be needed to detect parasites. A good sample – fresh not frozen- centrifuge method for the float, a giardia snap tests and cytology are the basics to evaluate the situation.
2. A thorough and accurate history of the diet is needed to rule out its involvement in the situation. Not what you meant to do but what actually happened with the diet and changes.

If the history clearly demonstrates that this particular animal only has diarrhea on or immediately after a run and all later bowel movements are normal, this can be initially treated as stress or exercise induced diarrhea. Stress diarrhea is not associated with vomiting or lack of appetite but often worsens as the ambient working temperature increases.

The following is a simple outline we use:

Try to feed these dogs 24 hours before they run to minimize stool in colon when they exercise.

Add one tsp of plain Metamucil per 30# of dog to the broth on the day of exercise. Do not add the Metamucil to each feeding and brothing. Beet pulp may have a similar beneficial affect.

Give a probiotic daily while the patient is running. Culterelle is a tested and proven probiotic. Be cautions when using supplement/vitamin products – these are not FDA regulated.

If the above does not work, a fecal stain for Clostridial overgrowth may be indicated. In addition there are several tests that are specialty tests that have to be sent to an outside lab. These tests often have very specific requirements to be accurate.

Often working dogs will be more predisposed to stress diarrhea after they have recovered from a "gut" bug. The intestinal lining can take up to 21 days (or longer in chronic cases) to fully recover.

Psyllium powder (water soluble fiber) on husks, Prostora (probiotic) are the gold standard created specifically for dogs.

Louse Infestations

Louse infestation (pediculosis) is usually seen during the winter or on young puppies. The rate of infestation is much higher in larger kennels, more mobile populations, or in severely debilitated animals.

Transmission: Lice are very species specific and will only live and feed on their correct host. The transmission must be direct, hair-to-hair. If off the intended host, lice will die within a few hours. Some dogs can be carriers. This means there are just a few lice that do not cause any symptoms on that host, but can infest other dogs that come in contact with this particular dog.

Lousy facts: There are two types, chewing and sucking lice. The chewing variety cause the dog to itch a little more and the sucking type can cause anemia if in large enough numbers. A louse-infested dog can have anything from a normal coat to being bald with open sores and a greasy, smelly coat. The animals are usually "itchy". Lice spend their entire time on the dog, making them much easier to eradicate than other insect parasites. From the time an egg is laid until the louse is an adult is 14 to 21 days.

Diagnosis: The best way to diagnose lice infestations is to look around the head, shoulders, and chest. Puppies that have "lots of scratches from playing with other puppies" should be closely examined. Lice can commonly be seen running across the bridge of the nose where the hair is very thin. They are much slower moving than fleas. If a litter of puppies "comes down" with lice, the female is infested and you have lice in your dog yard. Don't argue, it just is.

Treatment: Because lice cannot live off the host and they are sensitive to most insecticides, lice are easy to eliminate. Currently, there are newer flea products on the market that only require topical applications. Not all are FDA approved for lice so you may need to consult your veterinarian. (They may even repel mosquitoes!) Be sure to read the labels on these products as far as pregnant females and puppies are concerned. Follow the recommendations on the packaging for repeat treatments. To eliminate lice from your kennel you should treat all dogs and clean and disinfect all housing. Did you forget the dog truck?

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